

elling. A new option of automatic fixture removal is introduced that enables users to easily de-embed the test fixture/carrier plate used during device testing."

Another broader trend, which is good for both wafer-level users as well as normal analyser users, is that, spectrum analyser feature is now introduced in network analysers. This would enable users to carry out device-under-test analysis under modulated conditions or analyse spurious and harmonic responses of the device with a single connection, which will ease the measurement effort and overall test throughput.

Dwivedi adds, "Newer economy-range network analysers also feature direct receiver access that improves the system dynamic range. This is a good option for people looking for economical options for active-device-characterisation systems at lower frequencies."

Keysight has also launched low-cost network analysers for manufacturing lines as well as multi-port network analysers that work on modular platforms like PXI, which can be used for simultaneous testing of multiple devices for very fast measurements during production testing.

April Fang, sales manager at Siglent, explains, "Newer spectrum analysers include support for four independent traces and cursors, demodulation at zero span, advanced power measurement and a quasi-peak detector following CISPR 16."

Fang adds that newer instruments come with newer capabilities like importing stored limit templates that make it easy to evaluate electromagnetic interference levels for pre-compliance testing. Their SSA3000X includes a front-panel audio-output jack for listening to amplitude modulation and frequency modulation demodulated signals as well as a single-page menu that gives easy access to important measurement functions.

"I particularly like Tektro-

nix RSAA306 and our own Analog Arts SL987. These products are portable and offer full-featured spectrum-analysis capability," adds Shayan Ushani, editor, Analog Arts Inc.

Better usability

More firms have started unifying the user interface design across their product lines so as to help engineers work easier and faster when they switch between different equipment lines.

For instance, frequency upgradability lets customers buy equipment that can handle lower frequencies and then later upgrade to the feature set of higher-end equipment. While the upgrade is a software licence change in standard equipment, it is a hardware upgrade when making upgrades such as moving from 26GHz to 40GHz.

Test system requirements

When you build a system for RF testing, following are the instrument types that you would probably need:

- Signal generator
- Oscilloscope
- Connectors and cables
- Spectrum analyser
- Network analyser
- T&M software for automation
- Device under test

How instrument costs vary

India is becoming a hot market for high-technology products such as RF test equipment. Ushani says, "In India, just like in similar markets, cost of these products has always been a big factor in customers' decision-making criteria. Knowing this, manufacturers of such equipment have been trying to develop these instruments with consistently lower price tags."

Overall, vendors say that pricing of equipment remains mostly the same with the only exception being effects due to the fluctuating value of the rupee.

Major contributors to this report

- April Fang, sales manager, Siglent
- Ashutosh Dwivedi, application engineer, Keysight
- Dr Jan Benhelm, head of marketing, Zurich Instruments AG
- Shayan Ushani, editor, Analog Arts Inc.

Improved specifications

In signal generation, newer equipment focus on wideband signal generation with a maximum of 120MHz in the economy line of equipment and 160MHz in mid-range equipment. If you need more, high-end equipment can go up to 2GHz by using an external arbitrary waveform generator. These would be of use to engineers working to develop solutions for the defence market, electronic warfare testing (such as generating a threat environment).

High-end equipment feature an ultra-fast settling time. Keysight has some high-speed signal generators such as UGX X-series, which can cover from 10MHz to 40GHz.

Arbitrary waveform generator portfolio has also grown to use ultra-high bandwidth. While it was 5GHz earlier, newer equipment can handle up to 32GHz of analogue bandwidth. It can also handle up to 92 gigasamples per second. Predominantly used by guys in optical design who need high optical data rate, it can also be used for optical receiving testing and modulation like PAM-4, PAM8 and DMT. It also has an ultra-low jitter value so you get a very clean signal at such a high bandwidth and with a high data rate.

FieldFox is another product line that has gone up from 26.5GHz to cover up to 32GHz, 44GHz or even 50GHz range in the same handheld form factor. This is a battery-operated instrument useful for field testing, microwave network establishment and field testing of military radars working in microwave and mmWave frequency bands.

Looking forward to 2016

Going by the number of updated instruments, 2015 seems to have been quite the year for various RF test equipment. Vendors also talked about

a renewed focus on giving importance to the engineer using the equipment, through updates to the user interface, support software and productivity enhancing features. **EFY**